Architecting the Semantic Waste Layer: A Comprehensive Discovery of Multilingual Data Sources for the Fessi Knowledge Graph

Executive Summary

The development of the 'Fessi' knowledge graph represents a pivotal advancement in the domain of digital circular economy infrastructure. The central challenge in modern waste management is not merely logistical but semantic: bridging the chasm between the highly technical, regulated vocabulary of waste disposal (e.g., "composite packaging with aluminum barrier") and the colloquial, heterogeneous language used by consumers (e.g., "juice box," "tetra pak," "empty carton"). This challenge is compounded in multicultural societies like Germany, where waste sorting rules—*Mülltrennung*—are strictly enforced yet communicated primarily in German, creating a barrier for the significant Turkish and Arabic-speaking populations.

This report provides an exhaustive analysis of the data landscape required to populate Fessi. We have identified, analyzed, and categorized a diverse array of data sources that, when integrated, form a robust ontology capable of resolving multilingual synonyms, classifying ambiguous items, and navigating hyper-local disposal regulations. The analysis reveals that no single "master database" exists. Instead, Fessi must be constructed as a federated knowledge system that layers global semantic data (Wikidata, BabelNet) over regulatory thesauri (GEMET, EuroVoc), product-specific attributes (Open Food Facts, GS1), and hyper-local municipal rule sets (Abfallnavi, Abfall+).

Our research identifies six primary clusters of data sources:

1. **Linguistic Linked Open Data (LLOD):** Massive semantic networks like Wikidata and BabelNet that provide the multilingual synonymy and conceptual relationships necessary to parse natural language queries.
2. **Regulatory Environmental Thesauri:** Authoritative vocabularies such as GEMET and EuroVoc that define the legal boundaries of waste categories, essential for compliance and hazardous waste identification.
3. **Product and Material Registries:** Databases like Open Food Facts and the LUCID Packaging Register that bridge the gap between a branded product (e.g., "Nutella jar") and its constituent materials (Glass, Paper, PVC).
4. **Municipal Data Streams:** The fragmented but critical ecosystem of local APIs (Abfallnavi, MyMuell) and "Abfall ABC" lists that contain the ground-truth disposal rules

for specific jurisdictions.

5. **Hazardous Safety Protocols:** Chemical inventories (ECHA, GHS) that provide the logic for identifying and diverting dangerous household waste.
6. **Geospatial and Visual Context:** OpenStreetMap and computer vision datasets that ground abstract disposal concepts in physical locations and visual recognition.

The following report details the technical specifications, linguistic coverage, and integration strategies for each source, providing a blueprint for the Fessi engineering team to construct a system that is not only knowledgeable but culturally and legally attuned to the complexities of the German waste management landscape.

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1. The Semantic Backbone: Linguistic Linked Open Data (LLOD)

The primary requirement for Fessi is a "semantic skeleton"—a structural framework that connects abstract concepts across languages and domains. Linguistic Linked Open Data (LLOD) sources provide this infrastructure, offering stable identifiers (URIs) for concepts that allow the graph to treat "trash," "Müll," "atık," and "نفايات" not just as text strings, but as the same fundamental entity.

1.1 Wikidata: The Graph of Everything

Wikidata serves as the cornerstone for the Fessi knowledge graph due to its open license (CC0), massive community maintenance, and highly structured property system. Unlike a standard dictionary, Wikidata models the *relationships* between objects, materials, and disposal methods, making it indispensable for inferential reasoning.

1.1.1 Structural Properties for Waste Identification

The utility of Wikidata lies in its specific properties that describe materiality and recyclability. Our research highlights several key properties that Fessi must harvest to build its material reasoning engine.

Property P5994 (Recycling Code):
This property is the direct link between a material concept and its industrial identifier. The research explicitly identifies P5994 as the property "used to identify the material from which an item is made, to facilitate easier recycling or other reprocessing".1 In the context of the Fessi graph, P5994 serves as a critical normalizing agent. Whether a user queries "PET," "Plastic #1," or "Polyester," the graph can resolve these terms to the entity Q145626 (Polyethylene terephthalate) and retrieve the value "01" via P5994. This code is the key that unlocks the disposal logic for the "Gelber Sack" (Yellow Sack) system in Germany, which relies on these resin codes for sorting.

Property P186 (Material Used):
While P5994 identifies the code, P186 describes the physical composition of an object.1 This distinction is vital for answering object-oriented queries. If a user asks, "Can I recycle this chair?", Fessi can query the item "plastic chair" in Wikidata. If the P186 value links to "Polypropylene" (Q14592), the system can infer recyclability based on the material properties, even if the specific object "chair" isn't explicitly listed in a local waste guide. This allows for transitive inference: Object X is made of Material Y; Material Y is recyclable; therefore, Object X is potentially recyclable.
Property P279 (Subclass Of):
The hierarchical nature of waste requires a robust taxonomy. P279 builds this "Is-A" hierarchy. For instance, the entity "Wine Bottle" is a subclass of "Bottle," which is a subclass of "Container," which is a subclass of "Packaging".3 This hierarchy allows Fessi to apply general rules to specific instances. If the municipal rule states "All glass packaging goes in the bottle bank," the graph uses P279 to understand that a "Jam Jar" inherits the properties of "Glass Packaging" and should be disposed of accordingly. Conversely, it helps in exclusion logic: distinguishing "Window Glass" (Plate Glass) from "Container Glass" to prevent contamination, as they have different melting points.
Property P5997 (Object Stated in Reference As):
This property can be used to capture the specific terminology used in external references.1 This is particularly useful for tracking how different regulatory bodies refer to the same waste stream, creating a cross-reference system within the graph that can handle bureaucratic synonyms.

1.1.2 Multilingual Synonym Extraction Strategies

Wikidata's label and alias system is perhaps its most valuable asset for the multilingual requirement of Fessi. Each conceptual entity (Q-item) supports labels and aliases in hundreds of languages.

- **German (de):** The German dataset in Wikidata is exceptionally high-quality, often containing compound nouns that are specific to the DACH region (Germany, Austria, Switzerland). Terms like "Mülltrennung" (waste separation) or "Altglascontainer" (waste glass container) are often present as labels or aliases.[5] Extracting these allows Fessi to match the precise vocabulary found in municipal pamphlets.
- **Arabic (ar) and Turkish (tr):** The research indicates that Wikidata's coverage extends to these languages.[1] For a concept like "Recycling" (Q1647), Wikidata provides the Arabic label "إعادة تدوير" and the Turkish label "Geri dönüşüm." By harvesting the skos:altLabel (aliases) for these languages, Fessi can capture dialectal variations. For example, "Waste" might be labeled "نفايات" (Nufayat) but aliased as "قمامة" (Qumama) or "مخلفات" (Mukhallafat).[7] Capturing this synonymy is essential for a conversational agent that must understand the user's natural dialect, not just the formal standard language.

1.1.3 Query Implementation (SPARQL)

To populate Fessi, we recommend a series of federated SPARQL queries. The following logic

outlines the extraction strategy:

1. **Identify Core Nodes:** Select all entities that are subclasses (P279*) of "Waste" (Q45701), "Recycling" (Q1647), or "Packaging" (Q207514).
2. **Extract Material Attributes:** For these entities, retrieve the values for P5994 (Recycling Code) and P186 (Material Used).
3. **Harvest Multilingual Lexicon:** For every identified entity, extract the rdfs:label and skos:altLabel for the language codes en, de, ar, and tr.
4. **Filter for Relevance:** Use property P5995 (Kit Supplier) or other irrelevant properties found in snippets [1] as negative filters to ensure the graph is not polluted with sports data or film categories, which share some property ranges.

1.2 BabelNet: The Encyclopedic Semantic Network

While Wikidata provides the structural skeleton, BabelNet offers the flesh of lexicographic breadth. As a multilingual encyclopedic dictionary and semantic network, BabelNet integrates data from WordNet, Wikipedia, Wiktionary, and OmegaWiki, covering 500 languages.[8]

1.2.1 Synset Richness for Disambiguation

BabelNet organizes knowledge into "synsets" (sets of synonyms). This is crucial for handling polysemy—words with multiple meanings. In waste management, disambiguation is a safety-critical feature.

- **Case Study: "Glass":** The word "Glass" can refer to the material (recyclable), a drinking vessel (often not recyclable if crystal/lead), or a window pane (construction waste). BabelNet's domain categorization helps distinguish these senses.[10] By linking the user's query to the correct synset, Fessi can determine if the item belongs in the "Glass Container" or the "Residual Waste" bin.
- **Case Study: "Battery":** Fessi must distinguish between "Battery" (electrical accumulator - hazardous waste) and "Battery" (artillery unit - irrelevant). BabelNet's integration of Wikipedia categories and WordNet glosses provides the semantic context to filter these effectively.

1.2.2 Multilingual Alignment and Coverage

BabelNet's architecture is designed to align concepts across languages automatically. This is particularly valuable for bridging the gap between German and the target migrant languages.

- **Bridging the Gap:** If a specific waste term exists in the German Wikipedia (e.g., "Wertstoffhof" - Recycling Center) but lacks a direct article in the Arabic Wikipedia, BabelNet often infers the connection via inter-language links or shared WordNet parents. This allows Fessi to provide the Arabic user with the correct term for "Recycling Center" even if a direct dictionary translation is missing.[7]
- **Version Compatibility:** The research highlights BabelNet 5.3 (Java API) and Python API v1.1.0 as accessible integration points.[8] The availability of a local index (approx. 22M

entries) allows for low-latency querying, essential for a responsive user agent.

1.3 ConceptNet: Capturing "Folk Knowledge"

Regulators define waste by its material properties; people define it by its usage. ConceptNet excels at capturing this "common sense" or "folksonomy"—the way people actually speak and think about objects.[11]

1.3.1 Relational Edges for Waste Logic

ConceptNet's graph is built on a set of intuitive relations that map perfectly to waste sorting logic. Fessi can leverage these edges to infer disposal routes for items that are not explicitly listed in regulatory databases.

- **MadeOf:** This relation links objects to materials (e.g., /c/en/bottle MadeOf /c/en/plastic). This allows Fessi to apply material-level rules to object-level queries.
- **UsedFor:** This relation provides functional context (e.g., /c/en/newspaper UsedFor /c/en/reading). While less direct for disposal, it helps in identifying "contaminated" items. For example, if an object is UsedFor "cleaning toilets" (e.g., a brush), Fessi might infer it belongs in residual waste due to hygiene concerns, rather than recycling.
- **AtLocation:** This relation can help identify context. An item found AtLocation "kitchen" vs. "garage" might imply different waste streams (packaging vs. hazardous chemicals).

1.3.2 Linguistic Nuance and Granularity

ConceptNet is particularly strong in handling multilingual synonyms through its /r/Synonym edge.

- **Indonesian/Malay Distinction:** The research notes that ConceptNet v5.8 has improved the separation between Indonesian (id) and Malay (ms).[12] This attention to linguistic nuance suggests a high level of rigor in its multilingual mappings, giving confidence that the distinctions between, for example, Egyptian Arabic and Levantine Arabic (if represented) or Turkish and Azerbaijani would be handled with similar care.
- **API Access:** ConceptNet provides a straightforward REST API, making it easy to integrate into the Fessi pipeline for real-time synonym expansion.[11]

1.4 Open Multilingual Wordnet (OMW)

For deeper lexical coverage, the Open Multilingual Wordnet (OMW) offers a linked grid of wordnets for over 150 languages.[13]

1.4.1 The Collaborative Interlingual Index (CILI)

OMW Version 2 introduces the Collaborative Interlingual Index (CILI), which decouples concepts from the Princeton WordNet (English) ID structure.[13] This is a significant architectural advantage for Fessi.

- **Direct Mapping:** It facilitates direct mapping between German and Turkish concepts without pivoting through English, reducing "semantic drift." For example, a specific German waste term might have a direct equivalent in Turkish due to the large diaspora community, which might be lost if translated first to English and then to Turkish.
- **Domain-Specific Terminology:** Research indicates that linking OMW with domain-specific resources (like EuroTerm) enhances coverage for specialized fields like waste management.[14] This suggests that Fessi should not rely on OMW alone but use it as a linguistic bridge to connect more specialized glossaries.

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2. Regulatory and Environmental Thesauri: The Language of Law

While LLOD sources provide general language, waste management is a legal domain governed by strict definitions. A "battery" in common parlance might mean "AA cell," but in legal terms, it might be a "Category 1 Portable Accumulator." Regulatory thesauri provide the controlled vocabulary necessary for compliance.

2.1 GEMET: The General Multilingual Environmental Thesaurus

Maintained by the European Environment Agency (EEA), GEMET is the definitive source for environmental terminology in the EU.[7]

2.1.1 Semantic Precision and Scope

GEMET defines waste not just as a word, but as a concept with a rigorous scope.

- **Definition:** Concept 9041 ("Waste") is defined as "Material, often unusable, left over from any manufacturing, industrial, agricultural or other human process".[7]
- **Hierarchy:** It places waste in a hierarchy: Themes -> Wastes -> Types of Waste -> Municipal Waste. This hierarchy helps Fessi organize its knowledge graph logically. It ensures that "Municipal Waste" inherits the properties of "Waste" but is distinct from "Industrial Waste."
- **Scope Notes:** GEMET provides "Scope Notes" that explicitly state what is *included* and *excluded* from a term. This allows Fessi to answer complex user queries like "Is nuclear waste considered municipal waste?" (Answer: No, based on GEMET hierarchies).

2.1.2 Arabic and Turkish Coverage

Crucially for Fessi, GEMET supports 36 languages, including explicit support for Arabic and Turkish.[7]

- **Arabic Data:** The snippet reveals dual synonyms for "Waste": "مخلفات" (Mukhallafat) and "نفايات" (Nufayat). This covers both the concept of "residue/leftovers" and "refuse/trash," capturing important semantic nuances.

- **Turkish Data:** The term "atık" is provided as the standard equivalent.
- **Alignment:** By using GEMET IDs (e.g., Concept 9041), Fessi can guarantee that the German term "Abfall" is legally and conceptually equivalent to the Arabic "نفايات" within the system, regardless of colloquial variations.

2.2 EuroVoc and the List of Wastes (LoW)

EuroVoc is the multilingual thesaurus covering the activities of the EU, providing the vocabulary for European law.[16]

2.2.1 The European List of Wastes (LoW) Integration

The LoW defines 842 waste types with six-digit codes, structured into chapters.[16]

- **Chapter 20:** "Municipal Wastes" is the most relevant for Fessi's consumer-facing goals.
- **Hazardous Flags:** Entries in the LoW marked with an asterisk (*) are legally hazardous. Fessi can use this as a binary flag: IF LoW code has * THEN trigger "Hazardous Waste" warning.
- **Mapping Strategy:** Fessi should map user terms to LoW codes. For example, "Baby Wipes" -> LoW 20 03 01 "Mixed Municipal Waste." EuroVoc provides the linguistic labels for these codes in 24 languages, offering a standardized way to display waste categories to users.

2.2.2 Domain Filtering

EuroVoc organizes concepts into domains. This allows Fessi to filter out homonyms. If a user asks about "Oil," EuroVoc helps distinguish between "Cooking Oil" (Municipal Waste - Chapter 20) and "Engine Oil" (Hazardous Waste - Chapter 13). This domain awareness prevents dangerous advice (e.g., telling a user to compost engine oil).

2.3 IATE (Interactive Terminology for Europe)

IATE is the EU's inter-institutional terminology database, offering high-reliability translations used by professional translators.[18]

2.3.1 High-Fidelity Terminology

IATE allows for the download of domain-specific subsets (e.g., "Environment," "Waste Management") in TBX format.[20]

- **Quality Control:** Terms are rated by reliability (1-4 stars). Fessi should ingest only 3 and 4-star terms to ensure the highest quality of advice.
- **Phraseology:** IATE captures specific phrases like "Separate Collection" (*Getrennte Sammlung*) or "Bring System" (*Bringsystem*). These technical terms are often found in municipal waste guides but are rarely in standard dictionaries. Having these accurate translations in Arabic and Turkish allows Fessi to parse official city announcements and

explain them to users.

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3. The Product-Material Gap: Linking Brands to Bins

A major cognitive hurdle in waste sorting is the "Product-Material Gap." Users think in terms of products ("Yogurt pot," "Nutella jar"), while disposal rules are written in terms of materials ("Polystyrene," "White Glass"). Fessi must bridge this gap by linking products to their constituent materials.

3.1 Open Food Facts (OFF): The Packaging Taxonomy

Open Food Facts is a massive, crowdsourced database of food products that contains detailed packaging data.[21]

3.1.1 The "Global Packaging Taxonomy"

OFF maintains a highly structured taxonomy specifically for packaging components.[23] This is arguably the most valuable source for Fessi's product-level reasoning.

- **Granular Schema:** The database does not just say "Recyclable." It breaks packaging down into components: Shape, Material, and Recycling Instruction.
- **Nested Data Structure:** A single product can have multiple packaging units. Snippet [25] illustrates this perfectly: a box of cookies might be modeled as:
  1. **Unit 1:** Shape: Box | Material: Cardboard | Recycling: Recycle
  2. **Unit 2:** Shape: Tray | Material: Plastic | Recycling: Throw-away
  - **Implication:** This allows Fessi to provide sophisticated, granular advice: "Please separate the plastic tray from the cardboard box. Put the box in the Paper bin and the tray in the Yellow Sack."

3.1.2 Multilingual Tagging System

The taxonomy uses a system of prefixed tags (e.g., en:bottle, de:Flasche, fr:Bouteille).[24]

- **Inference:** While Arabic and Turkish tags might be less populated than English or French, the structured nature of the data allows for easy translation. Fessi can map the en:plastic-bottle tag to the Arabic ar:قنينة_بلاستيكية once, and instantly unlock packaging advice for millions of products.
- **Data Access:** The entire database is available as a daily JSONL export (approx. 4.8GB compressed).[26] This allows Fessi to ingest the data offline and index it for fast retrieval, rather than relying on rate-limited live API calls.

3.2 GS1 Global Product Classification (GPC)

GS1 manages the GTIN (barcode) system and the Global Product Classification (GPC)

standard.[28]

3.2.1 The "Brick" Categorization

GPC categorizes products into "Bricks" (e.g., "Milk/Cream - Shelf Stable").

- **Integration:** While GPC focuses on the product contents, recent initiatives ("Verified by GS1") are adding packaging attributes to support sustainability.
- **Strategic Value:** Access to GPC data (often via GDSN networks) would allow Fessi to classify products even if they are not in Open Food Facts. If a barcode resolves to the GPC Brick for "Aluminum Foil," Fessi knows the material is aluminum and can advise accordingly.

3.3 The LUCID Packaging Register and "Der Grüne Punkt"

In Germany, the *Verpackungsgesetz* (Packaging Act) requires brands to register their packaging.[30]

3.3.1 Brand-Level Compliance Data

The LUCID database, maintained by the ZSVR, is a public register of producers.[31]

- **Trust Signal:** While it doesn't list every SKU's material, it confirms if a brand is compliant with the "Dual System" (recycling). Fessi can use this to identify brands that are financially contributing to the recycling system.
- **Green Dot (Der Grüne Punkt):** Data from the Green Dot system helps define the boundary between "Sales Packaging" (Gelber Sack) and "Transport Packaging" (Paper/Cardboard bin). This distinction is vital for online retail packaging (e.g., Amazon boxes), which follows different rules than product packaging.[32]

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4. Hyper-Local Data Streams: The "Abfall ABC"

Germany’s waste management is federalist. Disposal rules vary significantly between municipalities. What is recyclable in Berlin (Wertstofftonne) might be residual waste in a rural Bavarian district. Fessi needs hyper-local data to be accurate.

4.1 The "Abfall ABC" Phenomenon

Every municipality publishes an "Abfall ABC"—an A-Z list of items and their disposal routes. These lists are the "Ground Truth" for local rules.

4.1.1 Fragmentation and Scraping

These lists are often fragmented across PDFs, HTML tables, and proprietary apps.

- **Berlin (BSR):** Offers detailed lists including tricky items like "Kassenzettel" (receipts).[33]

- **Munich (AWM):** Provides specific rules on glass colors (Blue glass -> Green bin).[34]
- **Friesland:** Explicitly states the rule for blue glass bottles.[35]
- **Strategy:** Fessi must employ targeted scrapers for the top 50 German cities to ingest these ABCs. The data should be stored as overrides: Concept: Receipt -> Default: Restmüll -> Override (Berlin): Restmüll -> Override (Region X): Paper (if applicable).

4.1.2 Semantic Conflicts and Resolution Rules

Our analysis of the snippets reveals several consistent "conflict zones" where Fessi requires specific logic:

- **The "Receipt" (Kassenzettel) Conflict:**
  - *Old Rule:* Thermal paper contains Bisphenol-A -> **Restmüll**.[36]
  - *New Rule:* "Blue4est" thermal paper (blue/grey) is eco-friendly -> **Altpapier**.[37]
  - *Fessi Logic:* Fessi cannot simply say "Receipts go in the bin." It must prompt the user: "Is the receipt white or blue-grey?" This nuance builds user trust.
- **The "Baking Paper" (Backpapier) Trap:**
  - *Common Misconception:* It looks like paper, so it goes in Paper or Bio bin.
  - *Reality:* It is silicone-coated -> **Restmüll**.[38] Fessi must proactively correct this misconception.
- **The "Pizza Box" Paradox:**
  - *Rule:* Cardboard is paper, but grease ruins paper recycling.
  - *German Standard:* Greasy boxes -> **Restmüll**.[40]
  - *Fessi Logic:* Ask "Is the box stained with oil or cheese?" If yes -> Restmüll. If no -> Altpapier.

4.2 Municipal APIs: Abfallnavi and Abfall+

Several service providers aggregate municipal data into APIs.

4.2.1 Abfallnavi API

This API serves many municipalities (e.g., Aachen, Nürnberg) and is relatively open.[42]

- **Endpoints:**
  - GET /orte (Locations): Resolves street addresses.
  - GET /fraktionen (Fractions): Returns the types of bins available at that address (e.g., "Bio," "Paper," "Residual").
  - GET /termine (Dates): Returns the collection calendar.
- **Data Value:** Knowing the *available fractions* is crucial. If a user asks "Where does the compost go?" but the API reports no "Bio" fraction for their address (common in city centers), Fessi can intelligently advise: "Your building does not have a bio bin; put it in the residual waste."

4.2.2 Abfall+ and MyMuell

These providers cover other regions.

- **Reliability:** The research notes that the "Abfall+" API can be unstable or lack documentation, with some integrations relying on reverse-engineering or facing outages (e.g., the Wermelskirchen cyber attack).[44]
- **Strategy:** Fessi should treat these APIs as "secondary" sources, caching their data where possible to mitigate uptime risks.

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5. Multilingual Integration: Bridging the Language Barrier

Germany is home to millions of Turkish and Arabic speakers. Effective waste management requires bridging the "Official German" to "Migrant Vernacular" gap.

5.1 Official Translations as Rosetta Stones

Municipalities like Landkreis Karlsruhe and Bremen publish flyers in Arabic and Turkish.[46] These are high-value sources for "Official Synonyms."

- **Data Extraction:**
  - **German:** *Gelber Sack* (Yellow Sack)
  - **Turkish:** *Ambalaj atıkları* (Packaging waste) [48]
  - **Arabic:** العبوات الخفيفة (Light packaging) [49]
- **Concept Mapping:** Fessi must map the *concept* of the bin (Yellow Sack) to these translated terms. This ensures that when a user asks about "Ambalaj atıkları," Fessi understands they are referring to the Yellow Sack system.

5.2 Addressing "False Friends" and Cultural Gaps

- **The "Pringles Dose" Problem:** A classic confusion point. It looks like a paper tube but is a composite of paper, foil, and metal.
  - *Search Term:* Users might search for "Cips kutusu" (Turkish) or "علبة بطاطس" (Arabic).
  - *Disposal:* In most of Germany, this goes in the **Gelber Sack** (as composite packaging), not the Paper bin.[50]
  - *Implication:* Fessi must recognize the specific shape/brand "Pringles" as a composite material, distinct from the generic concept of "Paper Can."
- **"Mülltrennung" as a Concept:** The term itself is culturally specific. Research suggests that for some users, the very concept of detailed separation needs explanation, not just translation.[51] Fessi should support "educational" queries (e.g., "Why do I have to separate trash?") using content from Handbook Germany.[53]

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6. Hazardous Waste & Safety: The "Problemstoff" Layer

Misclassifying hazardous waste is a critical failure mode. Fessi must prioritize safety.

6.1 ECHA C&L Inventory and GHS

The European Chemicals Agency (ECHA) manages the Classification and Labelling (C&L) Inventory.[54]

- **H-Statements:** The Globally Harmonized System (GHS) uses standardized codes (e.g., H410 "Toxic to aquatic life").
- **Multilingualism:** These codes are language-agnostic keys. H300 translates perfectly to "Lebensgefahr bei Verschlucken" (DE) or "Mortel en cas d'ingestion" (FR) via the ECHA-term database.[55]
- **Implementation:** Fessi should check product ingredients against ECHA data. If a product matches a substance with a high-risk H-code, Fessi must advise disposal at a "Schadstoffmobil" (Mobile Hazardous Waste Unit).

6.2 Visual Recognition of Hazard Symbols

Users often identify hazards by symbols (e.g., the "Dead Fish" or "Skull and Crossbones").

- **Data Source:** Wikidata and Wikimedia Commons host the official SVG pictograms for GHS.[56]
- **User Interaction:** Fessi can present these icons to the user ("Do you see this symbol on the bottle?") to confirm hazardous status without requiring the user to read complex chemical names.

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7. Geospatial and Visual Grounding

7.1 OpenStreetMap (OSM): Locating the Bin

OSM is the definitive source for locating physical waste infrastructure.[58]

- **Tagging Schema:**
  - amenity=recycling
  - recycling_type=container (for glass/paper banks)
  - recycling:glass=yes
  - recycling:glass:colour=mixed (or red/blue handling)
- **Data Value:** Fessi can answer "Where is the nearest glass bin?" by querying OSM nodes. It can also handle the "Blue Glass" rule geospatially: if the nearest bin is tagged recycling:glass:colour=mixed, Fessi can confidently tell the user to put the blue bottle there.

7.2 Computer Vision Datasets

To support photo-based queries, Fessi can utilize open datasets like **TrashNet** and **ZeroWaste**.[60]

- **Training Data:** These datasets contain thousands of labeled images of waste categories (cardboard, glass, metal, plastic).
- **Synthetic Data:** Recent initiatives use AI (like the Gemini API) to identify waste from photos.[62] Fessi could integrate a similar vision layer to identify the "Pringles can" or "Pizza box" visually and apply the correct sorting rule.

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8. Data Ingestion & Architecture Strategy

To build Fessi effectively, the disparate data sources must be layered into a coherent ontology.

8.1 The Layered Ontology Model

Fessi should be architected with a hierarchy of authority:

Layer	Source	Function	Authority Level
1. Semantic Base	Wikidata, BabelNet	Universal concepts, Synonyms	Low (Default)
2. Regulatory	GEMET, EuroVoc, LoW	Legal definitions, Hazards	Medium
3. Product	Open Food Facts, GS1	Item composition, Packaging	High (Item Specific)
4. Local Rules	Abfall APIs, ABCs	Disposal logic, Exceptions	**Highest (The Law)**

8.2 Conflict Resolution Logic

The system must implement a "Specific Overrides General" logic.

- **Scenario:** User has a "Pizza Box."
  - *Layer 1 (Wikidata):* Pizza Box is Cardboard.
  - *Layer 4 (Local Rule - Munich):* Pizza Box (Greasy) is Restmüll.
  - *Result:* Fessi applies the Munich rule.

- **Scenario:** User has "Blue Glass Bottle."
  - *Layer 1 (Wikidata):* Glass is recyclable.
  - *Layer 4 (Local Rule):* Blue Glass goes in **Green** Container.
  - *Result:* Fessi advises "Green Container" (not White or Brown).

8.3 Data Source Summary Table

Source Name	Type	Key Data Provided	Multilingual Support	Licensing
Wikidata	LLOD Graph	Material IDs, Recycling Codes, Aliases	Excellent (100+ langs)	CC0
GEMET	Thesaurus	Environmental definitions, Hierarchy	High (36 langs)	Open (EEA)
BabelNet	Semantic Net	Synonyms, Disambiguation	Excellent	Mixed/Paid
Open Food Facts	Product DB	Packaging composition, Recycling tags	High	ODbL
ECHA / GHS	Safety DB	Hazard statements, Chemical safety	High (EU langs)	Public
Abfallnavi	API	Local collection dates, Bin types	German	Proprietary/Open
OpenStreetMap	Geospatial	Bin locations, Accepted materials	High (Tags)	ODbL

Conclusion

Populating the Fessi knowledge graph is a task of semantic integration rather than data creation. The data exists—fragmented across global graphs, EU thesauri, product databases, and municipal PDFs. By constructing a federated system that layers these sources, Fessi can solve the dual challenges of the "Product-Material Gap" (mapping a jar to glass) and the "Language Gap" (mapping *Nufayat* to *Restmüll*).

The strategic recommendation is to build the core ontology on **Wikidata** and **GEMET**, enrich it with product data from **Open Food Facts**, and govern it with the hyper-local rules scraped from **Abfall ABCs** and **Abfall APIs**. This approach ensures that Fessi is not just a translation tool, but a legally accurate, safety-conscious, and culturally competent advisor for waste management in Germany.

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